CLIP & LLaVA Lab

Questions and Answers

Multimodal vision-language models: zero-shot classification, semantic image retrieval, visual question answering and hallucination analysis

Author	Siham Yamlahi Alami
Status	PhD Student
Laboratory	LabTIC
Institution	ENSA Tangier
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Part A — CLIP

Exercises 1 to 6

Exercise 1 — Installation and Model Loading

Q1. What is CLIP?

CLIP (Contrastive Language-Image Pre-training) is a multimodal AI model developed by OpenAI that learns to associate images with text descriptions.

Q2. What are its two main components?

CLIP consists of two main components:

- **Image Encoder:** extracts visual features from images.
- **Text Encoder:** extracts semantic features from text.

Q3. What is the role of *CLIPProcessor*?

CLIPProcessor preprocesses images and tokenizes text into the format required by CLIP.

Q4. What is the role of *CLIPModel*?

CLIPModel encodes images and text into a shared embedding space and computes their similarity.

Exercise 2 — Load an Image

Q5. What is the size of the image?

The image has a resolution of **550 × 550 pixels** (width × height).

Q6. What type of data is stored in the PIL image object?

The PIL image object stores pixel data. In RGB mode, each pixel contains three color values (Red, Green and Blue), typically ranging from 0 to 255.

Exercise 3 — Image/Text Similarity

Q7. Which description has the highest probability?

“a photo of a cat” has the highest probability, at **98.86%**.

Q8. Is it compatible with the actual image?

Yes. The image shows a cat, and CLIP correctly identifies it.

Q9. Why can CLIP compare an image with a sentence?

CLIP uses an image encoder and a text encoder to represent both images and sentences in a shared embedding space, allowing their similarity to be measured.

Q10. What does *softmax()* do?

softmax() converts similarity scores into relative probabilities between 0 and 1, ensuring their sum equals 1.

Exercise 4 — Zero-Shot Classification

Q11. What is zero-shot classification?

Zero-shot classification is the ability of a model to classify images into categories without being specifically trained on those categories.

Q12. Why is it not necessary to train a new classifier?

Because CLIP has already learned the relationship between images and text during pre-training. It can classify new images by comparing them with text descriptions of candidate classes.

Q13. What happens if the correct class is not included in the candidate classes?

CLIP will still select the most similar class among the available candidates, even if it is incorrect. This can lead to misleading predictions.

Exercise 5 — Evaluate CLIP

Q14. What is the obtained accuracy?

CLIP achieved **66.67% accuracy**, correctly classifying **10 out of 15 images**.

Q15. Which images are incorrectly classified?

Five images were incorrectly classified:

True class	Predicted class
Pizza	Horse
Burger	Horse
Fruits	Bird
Castle	Bird
City	Airplane

Q16. Give at least two possible reasons for the errors.

Possible explanations include:

- **Ambiguous visual features:** some images may share visual characteristics with other classes, leading to incorrect predictions.
- **Prompt limitations:** generic class descriptions may not adequately represent the visual content.
- **Closed-set classification:** CLIP must select one of the candidate classes, even if the correct category is missing.

  Note: in this experiment, the candidate list included all 11 categories, so missing candidate classes do not directly explain the five incorrect predictions.

Exercise 6 — Prompt Engineering

Q17. Are the similarity scores identical for all prompts?

No. The similarity scores vary depending on how the text prompt is formulated.

Q18. Which prompt achieves the highest similarity score?

“a photograph showing a cat” achieves the highest CLIP similarity score, **28.3018**.

Q19. What does this experiment demonstrate?

This experiment demonstrates that CLIP is sensitive to prompt wording. Different descriptions of the same object can produce different similarity scores, highlighting the importance of prompt engineering in multimodal models.

Part B — CLIP Image Retrieval

Exercises 7 and 8

Exercises 7 & 8 — Image Database and Text-to-Image Retrieval

Experimental setup

Parameter	Value
Image database	70 images
Embedding dimensions	512
Query	“a person playing football”
Retrieval method	Cosine similarity
Results displayed	Top-5 images

Q20. What is the most similar image?

The most similar image shows a man playing football, with a cosine similarity score of approximately **0.288**.

Q21. Is it semantically relevant?

Yes. The image directly matches the query “a person playing football.” The second-ranked image also shows a football player, confirming that CLIP retrieves semantically relevant results.

Q22. Why does this method not require image labels?

CLIP represents images and text in a shared embedding space. It retrieves images based on semantic similarity rather than predefined class labels, allowing retrieval without manually labeling the images.

Q23. Give one real-world application of semantic image retrieval.

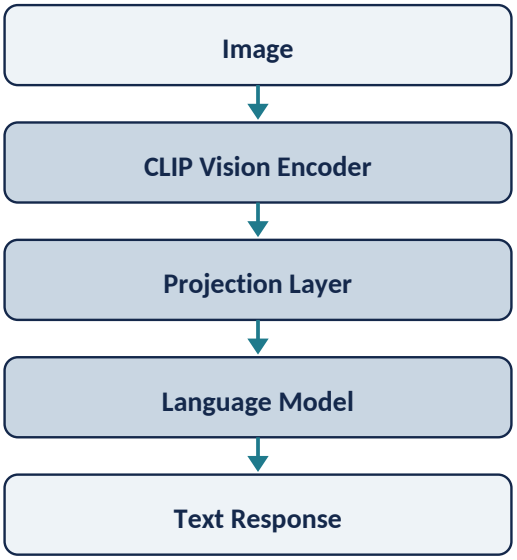
Image search engines: users can search large image collections using natural language descriptions, such as “a person playing football,” without manually tagging every image.

Part C — LLaVA

Exercises 9 to 12

Exercise 9 — Architecture

Completed architecture



Q24. What is the role of the vision encoder?

The vision encoder extracts visual features from an image and converts them into numerical representations (embeddings).

Q25. Why is a projection layer required?

The projection layer transforms visual embeddings into a representation compatible with the language model, allowing it to process visual information.

Q26. What is the role of the language model?

The language model processes the visual information together with the user's question to generate a natural language response.

Exercise 10 — Run LLaVA

Experimental setup

Parameter	Value
GPU	NVIDIA Tesla T4
GPU memory	14.56 GB
Model	LLaVA 1.5 7B
Quantization	4-bit NF4
GPU memory allocated after loading	4.65 GB

Observation

- LLaVA successfully processed an image of a hamburger and generated responses to five questions about the image.
- The model identified the hamburger, recognized that no people were visible, and described several ingredients.
- Some generated details, such as the cooking process, may not be directly supported by the image, illustrating the need to verify visual descriptions.

Exercise 11 — Visual Question Answering

Q27. What is the VQA accuracy?

LLaVA achieved **100% VQA accuracy**, correctly answering all **25 questions across 5 images (25/25)**, based on human evaluation.

Q28. Which types of questions are difficult for LLaVA?

Although no errors were observed in this experiment, LLaVA may struggle with:

- **Object counting:** accurately counting multiple objects in complex scenes.
- **Fine-grained details:** identifying small, partially hidden, or visually similar objects.
- **Spatial reasoning:** understanding precise relationships between objects.
- **Visual hallucinations:** generating plausible information that is not actually present in the image.

Note: the 100% accuracy reflects this small evaluation set and does not guarantee similar performance on more complex VQA tasks.

Exercise 12 — Hallucination

Q29. Does the bicycle actually exist?

No. The image shows a white airplane flying in the sky, and no bicycle is present.

Q30. Did LLaVA invent information?

Yes. Although LLaVA initially recognized that no bicycle was present, it later hallucinated two details:

- **Color:** it claimed the bicycle was white.
- **Location:** it claimed the bicycle was on the ground.

Q31. What is a visual hallucination?

A visual hallucination occurs when a multimodal model generates information about objects or details that are not actually present in the image.

Q32. Why can a multimodal model hallucinate?

Multimodal models generate responses based on learned patterns and visual representations. They may rely on assumptions suggested by the question instead of the actual image content, producing plausible but incorrect information.

> **Observation**
> LLaVA correctly identified the absence of the bicycle in the first response but invented its color and location in the following two responses.

Part D — CLIP vs LLaVA

Exercise 13

Exercise 13 — Comparative Experiment

Experimental setup: both models were tested on the same image of a horse.

CLIP results

- Predicted class: **Horse**
- Probability: **99.98%**
- CLIP correctly identified the horse among three candidate classes: cat, dog, and horse.

LLaVA results

- LLaVA identified a brown horse with a muscular build, a black mane, and a white stripe on its face.
- Unlike CLIP, LLaVA generated a detailed natural language description.

Property	CLIP	LLaVA
Input	Image + text descriptions	Image + text question
Main output	Similarity scores	Natural language response
Zero-shot classification	Yes	Yes, through prompting
Image retrieval	Yes	Not its primary purpose
Text generation	No	Yes
VQA	Limited	Yes
Open-ended response	No	Yes

Answers — Exercise 13

Q33. Which model is better suited to image classification?

CLIP is better suited to zero-shot image classification because it directly compares images with candidate text descriptions without requiring additional training.

Q34. Which model is better for image description?

LLaVA is better suited to image description because it generates detailed natural language descriptions of visual content.

Q35. Which model is better for visual question answering?

LLaVA is better suited to VQA because it can interpret images and generate answers to questions about their content.

Q36. Explain the fundamental difference between CLIP and LLaVA.

- **CLIP** is a contrastive vision-language model designed to align images and text through similarity scores, making it suitable for classification and retrieval.
- **LLaVA** is a generative vision-language model that combines a vision encoder with a language model to understand images and produce natural language responses.